Topology Optimization Lecture 4 - Constrained Optimization 1

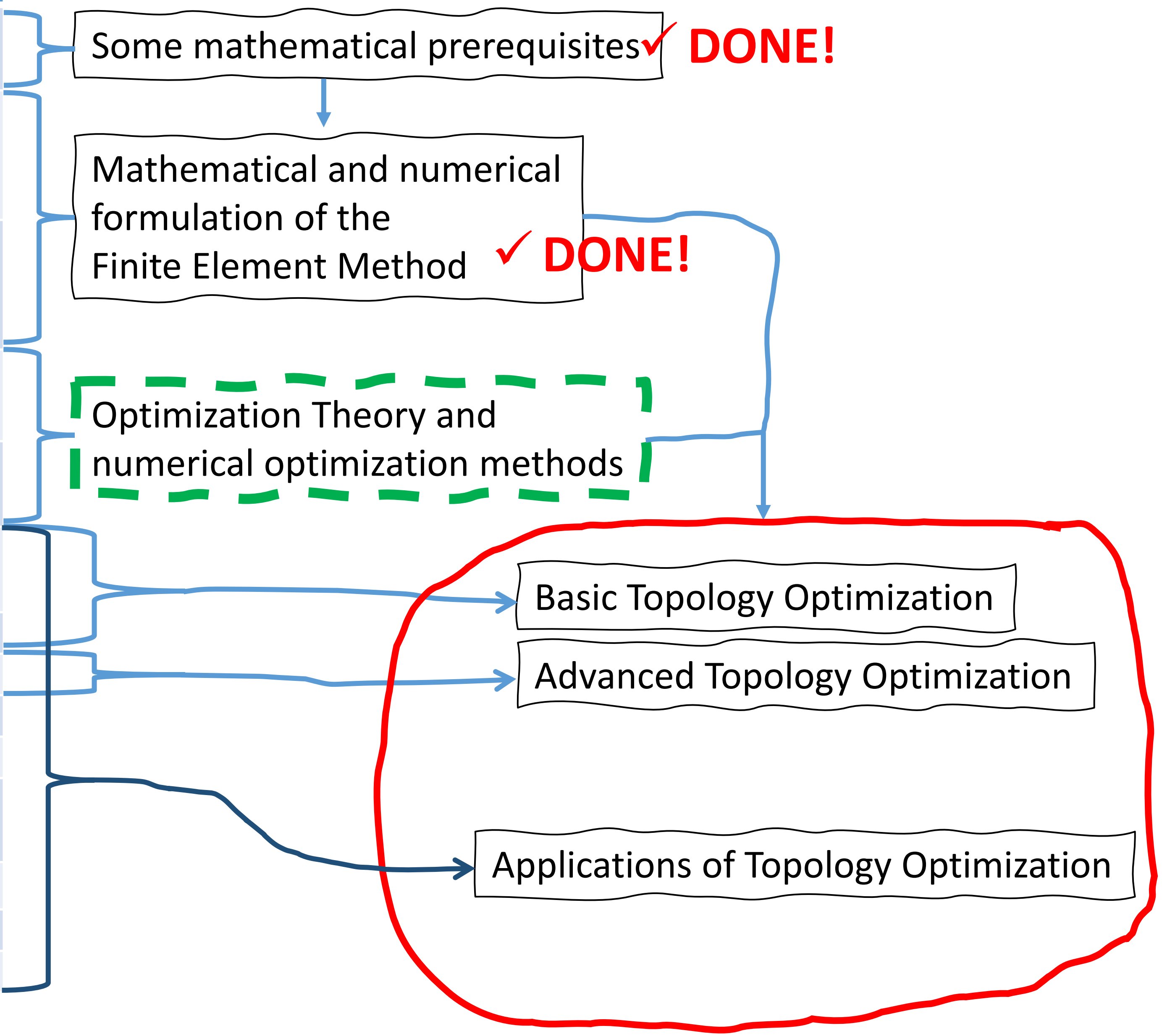
Optimality

KKT Theorem

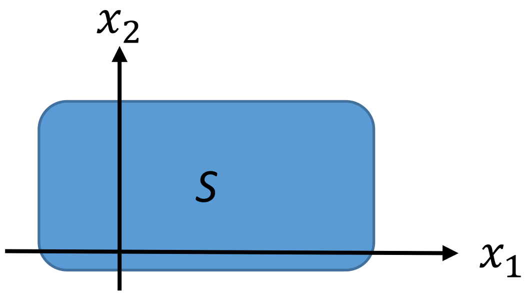
Sequential Linear Programming

Course overview

No.	Topics
1	Introduction Function and vector spaces, Interpolation, L^2 projection
2	The Finite Element Method for PDEs 1: Poisson's Equation, The Finite Element Method, Properties of the Finite Element Method
3	The Finite Element Method for PDEs 2: Index notation, Linear Elasticity, Non-linear Poisson's equation, Non-linear Elasticity
4	Constrained Optimization 1: Optimality, KKT Theorem, Numerical Optimization Algorithms
5	Constrained Optimization 2: Method of Moving Asymptotes, MPI and PETSc
6	Introduction to Topology Optimization: The idea, Filtering and Projection, Python implementation
7	Gradients
8	Robust topology optimization, More on MPI
9	Wave equations
10	Acoustic waves and more on Robust Optimization
11	Penalty and Augmented Lagrangian Methods Stress-constraints in Topology Optimization 1
12	Stress-constraints in Topology Optimization 2
13	Light concentration using nanoparticle design
14	Heat conduction



Global (absolute) Minimum:



A function $f(\boldsymbol{x})$ of n variables have a global minimum at $\boldsymbol{x}^*$ if

$$f(\boldsymbol{x}^*) \leq f(\boldsymbol{x}) \quad \forall \boldsymbol{x} \in S \text{ (feasible set)}$$

If strict inequality holds for all $\boldsymbol{x}$ other than $\boldsymbol{x}^*$, then $\boldsymbol{x}^*$ is a strong global minimum, otherwise it is a weak global minimum.

Local (relative) Minimum:

A function $f(\boldsymbol{x})$ of n variables have a local minimum at $\boldsymbol{x}^*$ if

$$f(\boldsymbol{x}^*) \leq f(\boldsymbol{x}) \quad \forall \boldsymbol{x} \in N(S, \boldsymbol{x}^*)$$
$$N(S, \boldsymbol{x}^*) = \{\boldsymbol{x} | \boldsymbol{x} \in S \text{ with } |\boldsymbol{x} - \boldsymbol{x}^*| < \delta\} \quad \text{for some small } \delta > 0$$

$N(S, \boldsymbol{x}^*)$ is the neighborhood of $\boldsymbol{x}^*$, i.e. the set of feasible points close to $\boldsymbol{x}^*$.
If strict inequality holds for all $\boldsymbol{x}$ other than $\boldsymbol{x}^*$, then $\boldsymbol{x}^*$ is a strong local minimum, otherwise it is a weak local minimum.

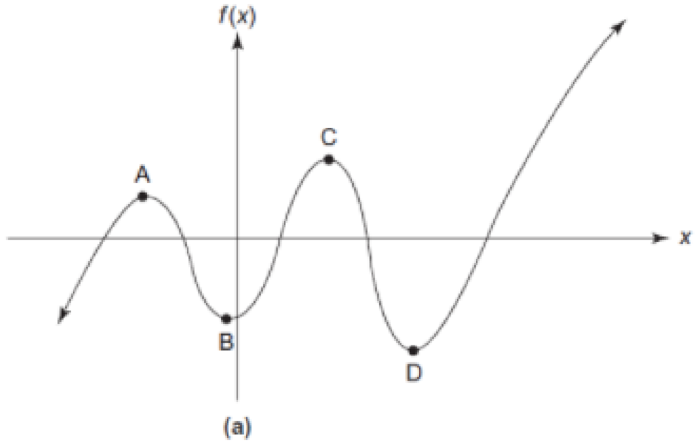
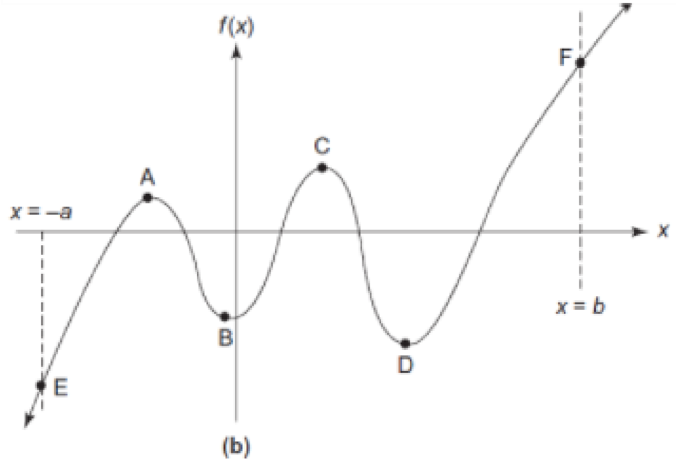


FIGURE 4.2 Representation of optimum points. (a) The unbounded domain and function (no global optimum). (b) The bounded domain and function (global minimum and maximum exist).



Necessary and Sufficient conditions

- Necessary conditions:
  - MUST be satisfied for a point to be an optimum point.
  - Points NOT satisfying necessary conditions are NOT optimum points
  - A point satisfying the necessary conditions is (only) a *candidate* optimum point.

- Sufficient conditions:
  - If a *candidate* point satisfies the sufficient conditions, then it IS an optimum point.
  - If a *candidate* point does not satisfy the sufficient conditions, then we don't know if it is an optimum point or not.

Unconstrained Optimization: One variable

Consider the Taylor expansion of a function of one variable:

$$f(x) = f(x^*) + f'(x^*)d + \frac{1}{2}f''(x^*)d^2 + R \quad \Rightarrow \quad \Delta f \equiv f(x) - f(x^*) \approx f'(x^*)d + \frac{1}{2}f''(x^*)d^2$$

with $d = x - x^*$. For a minimum point x^* we have $\Delta f \geq 0$ for all small d's.

The first order term dominates and d can be both positive and negative, so $\Delta f \geq 0 \Rightarrow f'(x^*) = 0$.

First-order necessary condition for x^* to be a minimum: $f'(x^*) = 0$.

One gets the same first-order necessary condition for x^* to be a maximum or saddle point.

For a candidate minimum point x^*:

$$\Delta f = f(x) - f(x^*) \approx f'(x^*)d + \frac{1}{2}f''(x^*)d^2 = \frac{1}{2}f''(x^*)d^2 \geq 0$$

Sufficient condition: Since $d^2 \geq 0$ then $f''(x^*) > 0$ must be satisfied for x^* to be a minimum.

2nd order necessary condition: $f''(x^*) \geq 0$. Any point not satisfying this is not a local minimum.

If both $f'(x^*) = 0$ and $f''(x^*) = 0$, a higher order Taylor expansion must be used in the analysis.

Unconstrained Optimization: One variable

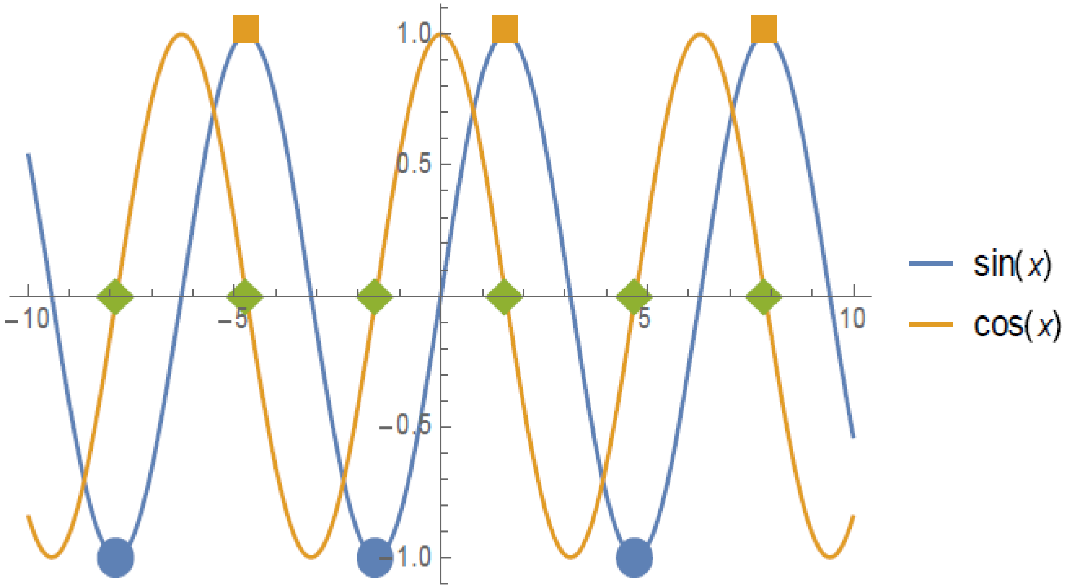
Find all minima and maxima of $f(x) = \sin x$.

Necessary condition: $f'(x^*) = 0 = \cos x^* \quad \Rightarrow \quad x^* = \pm\pi/2 + 2n\pi\ ,\ n \in \mathbb{Z}$ are candidate minimum points.

Sufficient condition for a minimum: $f''(x^*) = -\sin x^* > 0 \quad \Rightarrow \quad \sin x^* < 0$.

The candidate minimum points satisfying this sufficient condition are: $x^* = -\pi/2 + 2n\pi$. These are indeed the minimum points!

The remaining candidate minimum points are in fact maximum points, for which the sufficient condition is $f''(x^*) = -\sin x^* < 0 \quad \Rightarrow \quad \sin x^* > 0$.



Unconstrained Optimization: Many variables

For one variable:

$$f(x) = f(x^*) + f'(x^*)d + \frac{1}{2}f''(x^*)d^2 + R \quad \Rightarrow$$

$$\Delta f \equiv f(x) - f(x^*) \approx f'(x^*)d + \frac{1}{2}f''(x^*)d^2$$

The Taylor expansion of a function of many variables around $\boldsymbol{x}^*$ is:

$$f(\boldsymbol{x}) = f(\boldsymbol{x}^*) + (\nabla f(\boldsymbol{x}^*))^T \boldsymbol{d} + \frac{1}{2}\boldsymbol{d}^T \boldsymbol{H}(\boldsymbol{x}^*)\boldsymbol{d} + R \quad \Rightarrow$$

$$\Delta f = f(\boldsymbol{x}) - f(\boldsymbol{x}^*) = (\nabla f(\boldsymbol{x}^*))^T \boldsymbol{d} + \frac{1}{2}\boldsymbol{d}^T \boldsymbol{H}(\boldsymbol{x}^*)\boldsymbol{d} + R \quad , \quad \boldsymbol{d} = \boldsymbol{x} - \boldsymbol{x}^*$$

Necessary conditions: $\nabla f(\boldsymbol{x}^*) = \boldsymbol{0}$. $\boldsymbol{x}^*$ is a stationary point.

Sufficient conditions (minimum): $\boldsymbol{d}^T \boldsymbol{H}(\boldsymbol{x}^*)\boldsymbol{d} > 0$ for all $\boldsymbol{d}$, i.e. $\boldsymbol{H}$ must be positive definite.

Sufficient conditions (maximum): $\boldsymbol{d}^T \boldsymbol{H}(\boldsymbol{x}^*)\boldsymbol{d} < 0$ for all $\boldsymbol{d}$, i.e. $\boldsymbol{H}$ must be negative definite.

2nd order necessary conditions (minimum): $\boldsymbol{d}^T \boldsymbol{H}(\boldsymbol{x}^*)\boldsymbol{d} \geq 0$ for all $\boldsymbol{d}$, i.e. $\boldsymbol{H}$ must be positive semidefinite.

2nd order necessary conditions (maximum): $\boldsymbol{d}^T \boldsymbol{H}(\boldsymbol{x}^*)\boldsymbol{d} \leq 0$ for all $\boldsymbol{d}$, i.e. $\boldsymbol{H}$ must be negative semidefinite.

From the last two, we conclude that if $\boldsymbol{H}(\boldsymbol{x}^*)$ is indefinite, then the stationary point is not an optimum.

If $\boldsymbol{H}(\boldsymbol{x}^*)$ is indefinite and $\boldsymbol{x}^*$ is a stationary point, then $\boldsymbol{x}^*$ is termed an inflection point.

Unconstrained optimization algorithms

- There are many different algorithms for unconstrained optimization, often divided into 2 classes:
  - Derivative free algorithms.
    - Works often well for a small number of design variables.
  - Gradient based algorithms.
    - The class of algorithms to choose for ~>1000 design variables.
- Topology optimization requires most often A LOT of design variables, especially in 3D.
- Gradient based optimization is used a lot for topology optimization.
- Steepest descent is the simplest algorithm for gradient based optimization.
- It is not used a lot in practice but is the basis for understanding how many other algorithms work.
  - In machine learning it is often used in a stochastic version, which (sometimes) converge to the global optimum.
  - Sometimes it is used for topology optimization.
  - It can be used if other methods fail.
- For FEM problems, the required gradients can “easily” be calculated, as will be discussed in detail later in the course.

Steepest Descent

The most basic gradient-based method for choosing the descent direction is termed steepest descent or gradient descent.

The descent direction is choosen such that a small step, $\alpha^k > 0$, in the $\boldsymbol{d}^k$ direction, $\boldsymbol{x}^k + \alpha^k \boldsymbol{d}^k$, will decrease the cost function.

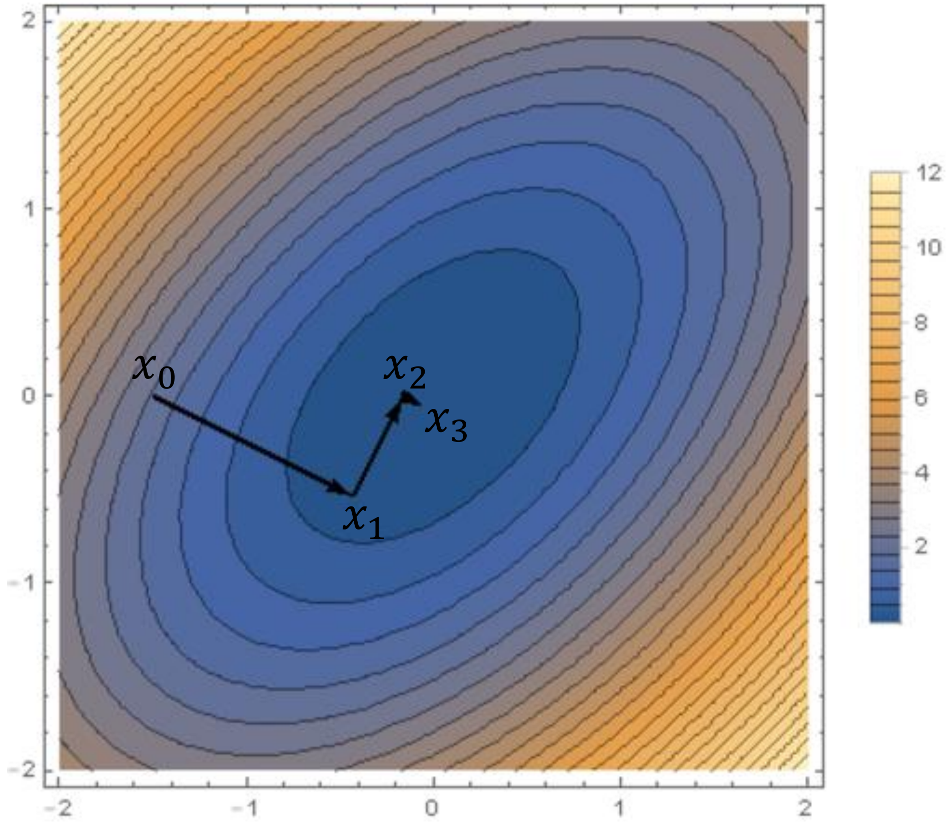
The *steepest descent direction* is $-\nabla f(\boldsymbol{x}^k)$ and

$$f(\boldsymbol{x}^k + \alpha^k \boldsymbol{d}^k) < f(\boldsymbol{x}^k) \quad \Rightarrow \quad f(\boldsymbol{x}^k) + \alpha^k \nabla f(\boldsymbol{x}^k) \cdot \boldsymbol{d}^k < f(\boldsymbol{x}^k) \quad \Rightarrow \quad \nabla f(\boldsymbol{x}^k) \cdot \boldsymbol{d}^k = -|\nabla f(\boldsymbol{x}^k)|^2 < 0$$

where a Taylor expansion and $\boldsymbol{d} = -\nabla f(\boldsymbol{x}^k)$ was used.

The algorithm is therefore:

Choose a starting point, $\boldsymbol{x}^0$.
Set $k = 0$.
While (Not converged) and $(k < k_{max})$
    Calculate $\nabla f(\boldsymbol{x}^k)$.
    $\boldsymbol{d}^k = -\nabla f(\boldsymbol{x}^k)$.
    Find α^k by minimizing $f(\boldsymbol{x}^k + \alpha^k \boldsymbol{d}^k)$.
    $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \alpha^k \boldsymbol{d}^k$.
    $k = k + 1$.



Convergence can be checked with e.g. $||\nabla f(\boldsymbol{x}^k)|| < \epsilon_1$ and/or monitoring the change in $\boldsymbol{x}$, $||\boldsymbol{x}^k - \boldsymbol{x}^{k-1}|| < \epsilon_2$.
ϵ_i are tolerances.

For the most very basic implementation, the step size α^k can be a constant or adjusted heuristically and the 1D minimization is not needed.

Constrained optimization

- Topology optimization (and other engineering optimization problems) often require constraints to be satisfied while minimizing a cost function.
- Some methods exist for applying unconstrained optimization algorithms to solve constrained optimization problems.
  - For example, adding penalty terms to the cost function, which have high values when the constraint is not satisfied.
- The preferred way is usually to consider the constraints directly while solving the constrained problem.
- We will study constrained optimization now.
- The most used algorithm in topology optimization, the method of moving asymptotes, will be discussed in detail next time.
- Penalty-like methods will be discussed later in the course for stress-based topology optimization.

Lagrange Multiplier Theorem

The basis for extending optimization methods to include constraints is the Lagrange multiplier theorem.

It conserns problems with equality constraints only.

The Lagrange Multiplier Theorem:

$$\min f(\boldsymbol{x}) \quad \text{subject to} \quad h_i(\boldsymbol{x}) = 0 \quad , \quad i = 1, \ldots, p$$

Let $\boldsymbol{x}^*$ be a regular point that is a local minimum for the problem. Then there exist p unique Lagrange multipliers v_j^*, $j = 1, \ldots, p$ such that

$$\frac{\partial f(\boldsymbol{x}^*)}{\partial x_i} + \sum_{j=1}^{p} v_j^* \frac{\partial h_j(\boldsymbol{x}^*)}{\partial x_i} = 0 \quad , \quad i = 1, \ldots, n$$

$$h_j(\boldsymbol{x}^*) = 0 \quad , \quad j = 1, \ldots, p$$

This can also be written using the Lagrange function L as:

$$L(\boldsymbol{x}, \boldsymbol{v}) = f(\boldsymbol{x}) + \sum_{j=1}^{p} v_j h_j(\boldsymbol{x}) = f(\boldsymbol{x}) + \boldsymbol{v}^T \boldsymbol{h}(\boldsymbol{x})$$

$$\frac{\partial L(\boldsymbol{x}^*, \boldsymbol{v}^*)}{\partial x_i} = 0 \ , \ i = 1, \ldots, n \ \Leftrightarrow \ \nabla L(\boldsymbol{x}^*, \boldsymbol{v}^*) = 0$$ (Gradient is only w.r.t. ***x***.)

$$\frac{\partial L(\boldsymbol{x}^*, \boldsymbol{v}^*)}{\partial v_j} = 0 \quad , \quad j = 1, \ldots, p$$

Lagrange Multiplier Theorem

$$L(\boldsymbol{x}, \boldsymbol{v}) = f(\boldsymbol{x}) + \sum_{j=1}^{p} v_j h_j(\boldsymbol{x}) = f(\boldsymbol{x}) + \boldsymbol{v}^T \boldsymbol{h}(\boldsymbol{x})$$

$$\nabla L(\boldsymbol{x}^*, \boldsymbol{v}^*) = 0$$

$$\frac{\partial L(\boldsymbol{x}^*, \boldsymbol{v}^*)}{\partial v_j} = 0 \quad , \quad j = 1, \ldots, p$$

The necessary conditions for a minimum of a cost function under equality constraints is found by looking at the gradient of the Lagrange function.

The gradient must be zero at an optimum and the equality constraints must be satisfied.

This is only necessary conditions, so it is also satisfied at maxima and saddle points.

If one makes an algorithm which always decreases the cost function until the Lagrnge Multiplier Theorem is satisfied, it will not locate a maximum.

Most often such an algorithm will also ”skip” saddle points and end up in a constrained minimum of the cost function.

In Topology Optimization, and other engineering optimization problems, inequality constraints are often used.

The Lagrange Multiplier Theorem must therefore be extended to include also inequalities.

Slack variables

We can turn an inequality constraint into an equivalent equality constraint by adding a slack variable:

$$g_i(\boldsymbol{x}) \leq 0 \quad \Leftrightarrow \quad g_i(\boldsymbol{x}) + s_i = 0 \;\; , \quad s_i \geq 0$$

If $s_i = 0$ then the constraint is satisfied at equality. The constraint is said to be *active*.

If $s_i > 0$ then the constraint is inactive.

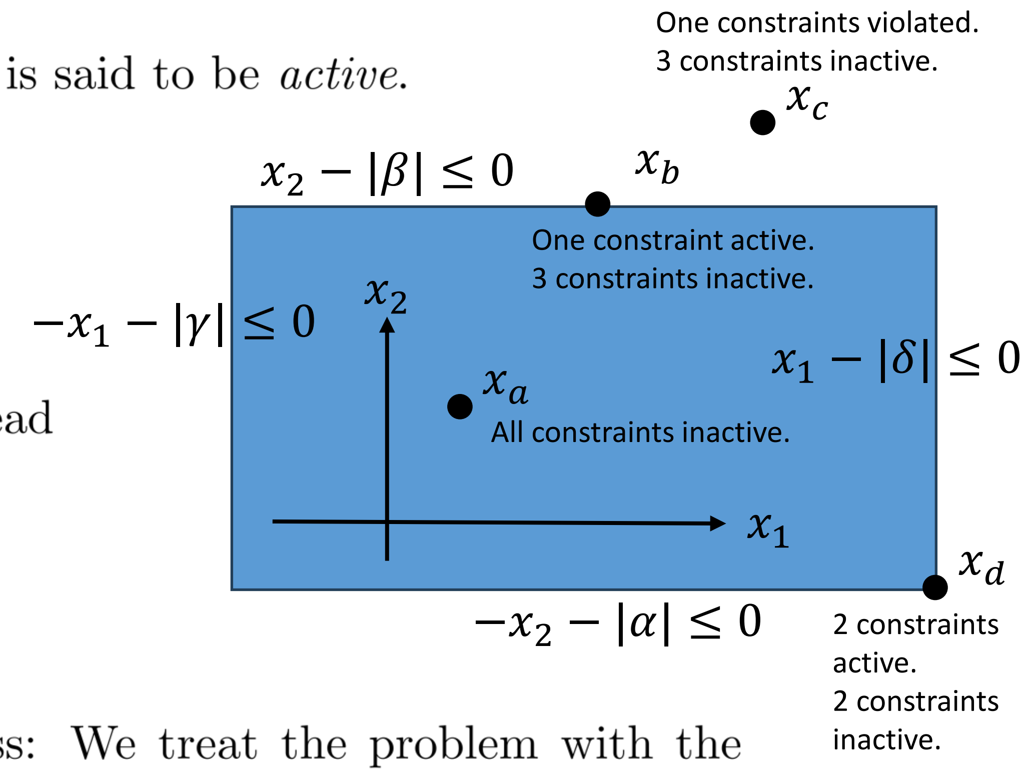
If $s_i < 0$ the constraint is violated.

We can get rid of the non-negativity constraint on s_i by using instead

$$g_i(\boldsymbol{x}) + s_i^2 = 0$$

If s_i is real, then $s_i^2 \geq 0$ automatically.

The value of s_i will be determined as part of the solution process: We treat the problem with the Lagrange multiplier theorem since it now only contains equality constraints, but with more variables - the slack variables.



THEOREM 4.6

Karush-Kuhn-Tucker Optimality Conditions

Let $\mathbf{x}^*$ be a regular point of the feasible set that is a local minimum for $f(\mathbf{x})$, subject to $h_i(\mathbf{x}) = 0$; $i = 1$ to p; $g_j(\mathbf{x}) \le 0$; $j = 1$ to m. Then there exist Lagrange multipliers $\mathbf{v}^*$ (a p-vector) and $\mathbf{u}^*$ (an m-vector) such that the Lagrangian function is stationary with respect to x_j, v_i, u_j, and s_j at the point $\mathbf{x}^*$.

1. Lagrangian function for the problem written in the standard form:

$$\begin{aligned} L(\mathbf{x},\mathbf{v},\mathbf{u},\mathbf{s}) &= f(\mathbf{x}) + \sum_{i=1}^{p} v_i h_i(\mathbf{x}) + \sum_{j=1}^{m} u_j (g_j(\mathbf{x}) + s_j^2) \\ &= f(\mathbf{x}) + \mathbf{v}^T \mathbf{h}(\mathbf{x}) + \mathbf{u}^T (\mathbf{g}(\mathbf{x}) + \mathbf{s}^2) \end{aligned} \tag{4.46}$$

2. Gradient conditions:

$$\frac{\partial L}{\partial x_k} = \frac{\partial f}{\partial x_k} + \sum_{i=1}^{p} v_i^* \frac{\partial h_i}{\partial x_k} + \sum_{j=1}^{m} u_j^* \frac{\partial g_j}{\partial x_k} = 0; \quad k = 1 \text{ to } n \tag{4.47}$$

$$\frac{\partial L}{\partial v_i} = 0 \Rightarrow h_i(\mathbf{x}^*) = 0; \quad i = 1 \text{ to } p \tag{4.48}$$

$$\frac{\partial L}{\partial u_j} = 0 \Rightarrow (g_j(\mathbf{x}^*) + s_j^2) = 0; \quad j = 1 \text{ to } m \tag{4.49}$$

3. Feasibility check for inequalities:

$$s_j^2 \ge 0; \text{ or equivalently } g_j \le 0; \quad j = 1 \text{ to } m \tag{4.50}$$

4. Switching conditions:

$$\frac{\partial L}{\partial s_j} = 0 \Rightarrow 2u_j^* s_j = 0; \quad j = 1 \text{ to } m \tag{4.51}$$

5. Nonnegativity of Lagrange multipliers for inequalities:

$$u_j^* \ge 0; \quad j = 1 \text{ to } m \tag{4.52}$$

6. *Regularity check*: Gradients of the active constraints must be linearly independent. In such a case the Lagrange multipliers for the constraints are unique.

Example

Minimize

$$f(\boldsymbol{x}) = (x_1 - 1.5)^2 + (x_2 - 1.5)^2 \quad \text{subject to} \quad g(\boldsymbol{x}) = x_1 + x_2 - 2 \leq 0$$

We first write down the Lagrange function for the problem:

$$\begin{aligned} L &= f(\boldsymbol{x}) + \sum_{i=1}^{p} v_i h_i(\boldsymbol{x}) + \sum_{j=1}^{m} u_j \left(g_j(\boldsymbol{x}) + s_j^2 \right) \\ &= (x_1 - 1.5)^2 + (x_2 - 1.5)^2 + u \left(x_1 + x_2 - 2 + s^2 \right) \end{aligned}$$

The gradient conditions are

$$\begin{aligned} \frac{\partial L}{\partial x_1} &= 2(x_1 - 1.5) + u = 0 \\ \frac{\partial L}{\partial x_2} &= 2(x_2 - 1.5) + u = 0 \\ \frac{\partial L}{\partial u} &= x_1 + x_2 - 2 + s^2 = 0 \end{aligned}$$

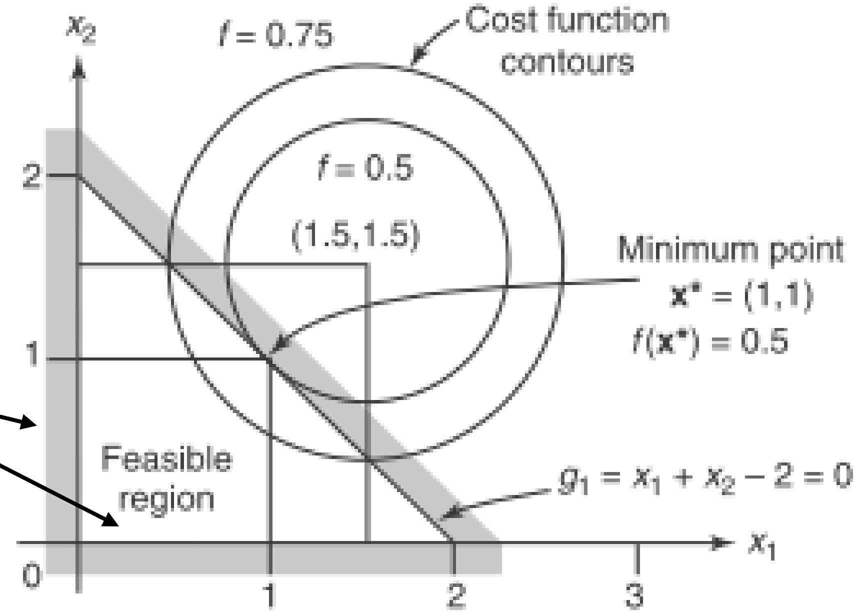
and the switching condition is

$$\frac{\partial L}{\partial s} = 2us = 0$$

Example

$$\frac{\partial L}{\partial x_1} = 2(x_1 - 1.5) + u = 0$$
$$\frac{\partial L}{\partial x_2} = 2(x_2 - 1.5) + u = 0$$
$$\frac{\partial L}{\partial u} = x_1 + x_2 - 2 + s^2 = 0$$
$$\frac{\partial L}{\partial s} = 2us = 0$$

These two constraints are *not* included in this example...



The last condition is the switching condition and is satisfied if $s = 0$, $u = 0$ or $s = u = 0$. The last case is termed abnormal and is usually not considered as it rarely give additional candidate points.

Solving the 4 eqs. with 4 unknown give:

Solution 1: $x_1 = x_2 = u = 1$ and $s = 0$. This is a candidate minimum point. From the graphical solution, we know it is in fact the minimum point.

Solution 2 and 3: $x_1 = x_2 = 1.5$, $u = 0$ and $s = \pm i$. This case does not satisfy the feasibility check $s^2 \geq 0$ and is thus not a candidate minimum point.

```
from sympy.solvers import solve
from sympy import symbols

x1, x2, u, s = symbols('x1 x2 u s')
sol=solve([2*(x1-1.5)+u , 2*(x2-1.5)+u , x1+x2-2+s**2 , 2*u*s])
print(sol)
----- output -----
[{s: 0.0, u: 1.00000000000000, x1: 1.00000000000000, x2:
1.00000000000000}, {s: -1.0*I, u: 0.0, x1: 1.50000000000000,
x2: 1.50000000000000}, {s: 1.0*I, u: 0.0, x1: 1.50000000000000,
x2: 1.50000000000000}]
```

Some notes on the KKT conditions

If $s_j = 0$, the corresponding inequality constraint is active.
If $s_j^2 > 0$, the constraint is inactive.

The switching conditions essentially switches inequality constraints from active to inactive and forces us to consider all combinations of this.

With m inequality constraints we get m switching conditions, resulting in 2^m normal cases. This grows quickly with the number of inequality constraints and the KKT conditions can not be solved in a reasonable time for a large number of inequality constraints.

We may find several candidate minimum points for each switching condition, if the remaining system is non-linear.

The solution of the KKT conditions goes like this:

1. Gradient and switching conditions give $n + 2m + p$ equations with $n + 2m + p$ unknown.
2. Solve these non-linear equations (the switching eqs. are always non-linear, the rest may be either linear or non-linear.).
3. For each solution, check feasibility as well as non-negativity of Lagrange multipliers for inequality constraints.
4. The points that pass the test are candidate minimum points.

Regular points that satisfy KKT conditions are candidate minimum points only!

The regular points that does not satisfy KKT are *not* minimum points.

THEOREM 4.6

Karush-Kuhn-Tucker Optimality Conditions

Let $\mathbf{x}^*$ be a regular point of the feasible set that is a local minimum for $f(\mathbf{x})$, subject to $h_i(\mathbf{x}) = 0$; $i = 1$ to p; $g_j(\mathbf{x}) \le 0$; $j = 1$ to m. Then there exist Lagrange multipliers $\mathbf{v}^*$ (a p-vector) and $\mathbf{u}^*$ (an m-vector) such that the Lagrangian function is stationary with respect to x_j, v_i, u_j, and s_j at the point $\mathbf{x}^*$.

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2. Gradient conditions:

$$\frac{\partial L}{\partial x_k} = \frac{\partial f}{\partial x_k} + \sum_{i=1}^{p} v_i^* \frac{\partial h_i}{\partial x_k} + \sum_{j=1}^{m} u_j^* \frac{\partial g_j}{\partial x_k} = 0; \quad k = 1 \text{ to } n \tag{4.47}$$

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$$\frac{\partial L}{\partial s_j} = 0 \Rightarrow 2u_j^* s_j = 0; \quad j = 1 \text{ to } m \tag{4.51}$$

5. Nonnegativity of Lagrange multipliers for inequalities:

$$u_j^* \ge 0; \quad j = 1 \text{ to } m \tag{4.52}$$

6. *Regularity check*: Gradients of the active constraints must be linearly independent. In such a case the Lagrange multipliers for the constraints are unique.

Scaling of cost function

If we solve two problems with the same constraints but different cost functions, f and Kf with $K > 0$, the minima will be the same.

The value of the cost function and the Lagrange multipliers at minima will, however, change.

The scaling is:

$$
\begin{aligned}
f^*_{\text{scaled}} &= Kf^* \\
v^*_{i\ \text{scaled}} &= Kv^*_i \\
u^*_{j\ \text{scaled}} &= Ku^*_j
\end{aligned}
$$

This can be seen from the KKT conditions.

THEOREM 4.6

Karush-Kuhn-Tucker Optimality Conditions

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1. Lagrangian function for the problem written in the standard form:

$$
\begin{aligned}
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&= f(\mathbf{x}) + \mathbf{v}^T \mathbf{h}(\mathbf{x}) + \mathbf{u}^T (\mathbf{g}(\mathbf{x}) + \mathbf{s}^2)
\end{aligned}
\tag{4.46}
$$

2. Gradient conditions:

$$\frac{\partial L}{\partial x_k} = \frac{\partial f}{\partial x_k} + \sum_{i=1}^{p} v_i^* \frac{\partial h_i}{\partial x_k} + \sum_{j=1}^{m} u_j^* \frac{\partial g_j}{\partial x_k} = 0; \quad k = 1 \text{ to } n \tag{4.47}$$

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6. *Regularity check*: Gradients of the active constraints must be linearly independent. In such a case the Lagrange multipliers for the constraints are unique.

Scaling of constraint functions

If we scale equality constraint functions by non-zero constants, P_i, and inequality constraint functions by positive constants, M_j,

$$h_{i,\text{scaled}} = h_i P_i \quad , \quad P_i \neq 0$$
$$g_{j,\text{scaled}} = g_j M_j \quad , \quad M_j > 0$$

neither the feasible set nor the cost function and the minima change.

The Lagrange multipliers change as

$$v^*_{i,\text{scaled}} = v^*_i / P_i$$
$$u^*_{j,\text{scaled}} = u^*_j / M_j$$

The slack variables change as

$$s^*_{j,\text{scaled}} = s^*_j \sqrt{M_j}$$

This can be seen from the KKT conditions.

THEOREM 4.6

Karush-Kuhn-Tucker Optimality Conditions

Let $\mathbf{x}^*$ be a regular point of the feasible set that is a local minimum for $f(\mathbf{x})$, subject to $h_i(\mathbf{x}) = 0$; $i = 1$ to p; $g_j(\mathbf{x}) \le 0$; $j = 1$ to m. Then there exist Lagrange multipliers $\mathbf{v}^*$ (a p-vector) and $\mathbf{u}^*$ (an m-vector) such that the Lagrangian function is stationary with respect to x_j, v_i, u_j, and s_j at the point $\mathbf{x}^*$.

1. Lagrangian function for the problem written in the standard form:

$$\begin{aligned} L(\mathbf{x},\mathbf{v},\mathbf{u},\mathbf{s}) &= f(\mathbf{x}) + \sum_{i=1}^{p} v_i h_i(\mathbf{x}) + \sum_{j=1}^{m} u_j (g_j(\mathbf{x}) + s_j^2) \\ &= f(\mathbf{x}) + \mathbf{v}^T \mathbf{h}(\mathbf{x}) + \mathbf{u}^T (\mathbf{g}(\mathbf{x}) + \mathbf{s}^2) \end{aligned} \tag{4.46}$$

2. Gradient conditions:

$$\frac{\partial L}{\partial x_k} = \frac{\partial f}{\partial x_k} + \sum_{i=1}^{p} v_i^* \frac{\partial h_i}{\partial x_k} + \sum_{j=1}^{m} u_j^* \frac{\partial g_j}{\partial x_k} = 0; \quad k = 1 \text{ to } n \tag{4.47}$$

$$\frac{\partial L}{\partial v_i} = 0 \Rightarrow h_i(\mathbf{x}^*) = 0; \quad i = 1 \text{ to } p \tag{4.48}$$

$$\frac{\partial L}{\partial u_j} = 0 \Rightarrow (g_j(\mathbf{x}^*) + s_j^2) = 0; \quad j = 1 \text{ to } m \tag{4.49}$$

3. Feasibility check for inequalities:

$$s_j^2 \ge 0; \text{ or equivalently } g_j \le 0; \quad j = 1 \text{ to } m \tag{4.50}$$

4. Switching conditions:

$$\frac{\partial L}{\partial s_j} = 0 \Rightarrow 2u_j^* s_j = 0; \quad j = 1 \text{ to } m \tag{4.51}$$

5. Nonnegativity of Lagrange multipliers for inequalities:

$$u_j^* \ge 0; \quad j = 1 \text{ to } m \tag{4.52}$$

6. *Regularity check*: Gradients of the active constraints must be linearly independent. In such a case the Lagrange multipliers for the constraints are unique.

Alternative form of KKT conditions

TABLE 5.1 Alternate form of KKT necessary conditions

Problem: Minimize $f(x)$ subject to $h_i(x) = 0$, $i = 1$ to p; $g_j(x) \le 0$, $j = 1$ to m		
1. Lagrangian function definition	$L = f + \sum_{i=1}^{p} v_i h_i + \sum_{j=1}^{m} u_j g_j$	(5.1)
2. Gradient conditions	$\dfrac{\partial L}{\partial x_k} = 0; \quad \dfrac{\partial f}{\partial x_k} + \sum_{i=1}^{p} v_i^* \dfrac{\partial h_i}{\partial x_k} + \sum_{j=1}^{m} u_j^* \dfrac{\partial g_j}{\partial x_k} = 0; \quad k = 1 \text{ to } n$	(5.2)
3. Feasibility check	$h_i(\mathbf{x}^*) = 0; \quad i = 1 \text{ to } p; \quad g_j(\mathbf{x}^*) \le 0; \quad j = 1 \text{ to } m$	(5.3)
4. Switching conditions	$u_j^* g_j(\mathbf{x}^*) = 0; \quad j = 1 \text{ to } m$	(5.4)
5. Non-negativity of Lagrange multipliers for inequalities	$u_j^* \ge 0; \quad j = 1 \text{ to } m$	(5.5)
6. Regularity check: Gradients of active constraints must be linearly independent. In such a case, the Lagrange multipliers for the constraints are unique.		

The slack variables have been eleminated in (4.51) using (4.49), $g_j(\boldsymbol{x}^*) = -s_j^2$, and one works directly with the inequality constraint functions, $g_j(\boldsymbol{x}^*)$, in the feasibility check.

Second order conditions for constrained optimization

(extremely briefly)

KKT conditions are only necessary conditions for a minimum and does not by themselves garantee a minimum. To garantee a minimum, a candidate point $\boldsymbol{x}^*$ must additionally satisfy sufficient conditions.

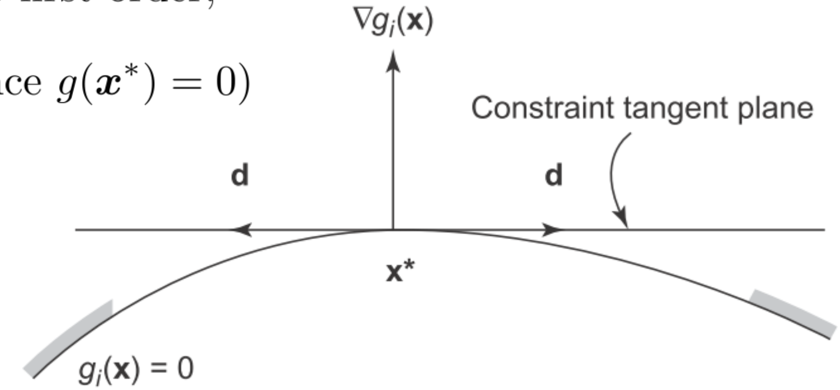
If there are no active constraints at $\boldsymbol{x}^*$, then any point close to $\boldsymbol{x}^*$ is in the neighborhood of $\boldsymbol{x}^*$, i.e. $N(S, \boldsymbol{x}^*) = \{\boldsymbol{x} | \boldsymbol{x}^* + \boldsymbol{d} \text{ with } |\boldsymbol{d}| << 1\}$.

What if we do have active constraints? Then $\boldsymbol{x}^* + \boldsymbol{d}$ must satisfy the active constraints to first order,

$$g(\boldsymbol{x}^* + \boldsymbol{d}) \approx g(\boldsymbol{x}^*) + (\nabla g(\boldsymbol{x}^*))^T \boldsymbol{d} = 0 \quad \Rightarrow \quad (\nabla g(\boldsymbol{x}^*))^T \boldsymbol{d} = 0 \quad (\text{since } g(\boldsymbol{x}^*) = 0)$$

Thus we get a set of equations to determine feasible moves away from $\boldsymbol{x}^*$:

$$\begin{aligned} (\nabla h_i(\boldsymbol{x}^*))^T \boldsymbol{d} = 0 \quad &, \quad i = 1, ..., p \\ (\nabla g_j(\boldsymbol{x}^*))^T \boldsymbol{d} = 0 \quad &, \quad \text{all active inequalities} \end{aligned}$$



The $\boldsymbol{d}$'s satisfying these conditions are in the constraint gradient tangent hyperplane of all active constraints.

To derive sufficient condition for a minimum, one can use a Taylor expansion of the Lagrange function around a candidate minimum point $\boldsymbol{x}^*$:

$$\begin{aligned} L(\boldsymbol{x}^* + \boldsymbol{d}) &\approx L(\boldsymbol{x}^*) + (\nabla L(\boldsymbol{x}^*))^T \boldsymbol{d} + \frac{1}{2}\boldsymbol{d}^T \nabla^2 L(\boldsymbol{x}^*) \boldsymbol{d} \\ &= L(\boldsymbol{x}^*) + \frac{1}{2}\boldsymbol{d}^T \nabla^2 L(\boldsymbol{x}^*) \boldsymbol{d} > L(\boldsymbol{x}^*) \quad \Leftrightarrow \quad \boldsymbol{d}^T \nabla^2 L(\boldsymbol{x}^*) \boldsymbol{d} > 0 \end{aligned}$$

To be a constrained minimum, this must be satisfied for all feasible directions $\boldsymbol{d} \neq 0$.

Global Optimality

- Most methods guarantee only a local optimum solution.
- By calculating all candidate minimum points, we can compare the cost function value for the candidate optimal points and find the global minimum (or minima).
- This is termed an *exhaustive search*.
- This is generally a very hard problem as the number of KKT points grow quickly with number of inequalities and the gradient-equations may be hard to solve.
- For a so-called *convex problem*, any local minimum is also a global minimum, which is a very handy feature!
- Most often in Topology Optimization, problems are non-linear and non-convex, so there may be multiple solutions, and the global optimum is not guaranteed to be found.
- Additional properties, such as physical bounds can sometimes be used to judge if the solution is close to a global optimum or not.

Constrained Optimization Algorithms

- We will now and next time discuss numerical algorithms for problems with inequality constraints.
- The most basic algorithm is Sequential Linear Programming.
- It should not be used for serious work, but the ideas are central to discussing much better algorithms.
- Good algorithms should be *globally convergent*, meaning that they will converge to a local minimum no-matter the starting point.
  - This does NOT mean they converge to the global minimum! ☹

Linearization

An often used technique is to linearize a non-linear problem and then solve a linear subproblem in each iteration.

Minimize $f(\boldsymbol{x})$ subject to $h_i(\boldsymbol{x}) = 0$ and $g_j(\boldsymbol{x}) \leq 0$ is at iteration k of an algorithm turned into the subproblem:

$$\begin{aligned}\text{Minimize } & f(\boldsymbol{x}^k) + \left(\nabla f(\boldsymbol{x}^k)\right)^T \boldsymbol{d}^k \\ \text{subject to } & h_i(\boldsymbol{x}^k) + \left(\nabla h_i(\boldsymbol{x}^k)\right)^T \boldsymbol{d}^k = 0 \ , \ i = 1, ..., p \text{ and} \\ & g_j(\boldsymbol{x}^k) + \left(\nabla g_j(\boldsymbol{x}^k)\right)^T \boldsymbol{d}^k \leq 0 \ , \ j = 1, ..., m\end{aligned}$$

$\boldsymbol{d} = \boldsymbol{x} - \boldsymbol{x}^k$ is the move away from the current point $\boldsymbol{x}^k$.

Example

Minimize $f(\boldsymbol{x}) = x_1^2 + x_2^2 - 3x_1x_2$ subject to $g_1(\boldsymbol{x}) = \frac{1}{6}\left(x_1^2 + x_2^2\right) - 1 \leq 0$, $g_2(\boldsymbol{x}) = -x_1 \leq 0$, and $g_3(\boldsymbol{x}) = -x_2 \leq 0$.

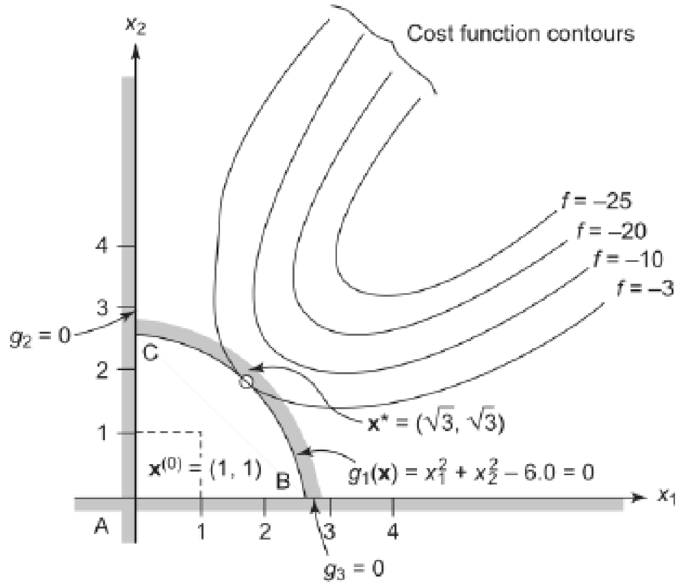
Linearized subproblem at $\boldsymbol{x}^k = (1,1)$: $\boldsymbol{d} = \boldsymbol{x} - (1,1)^T$ and

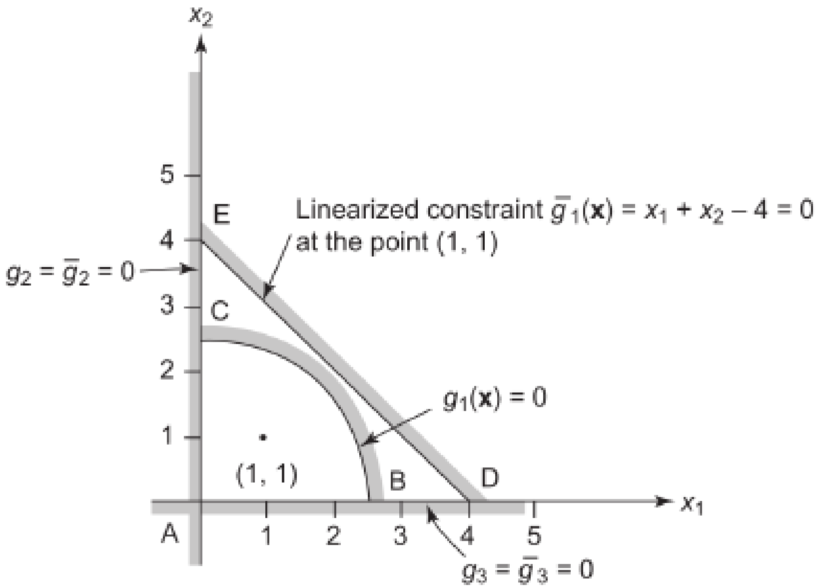
$$\text{Minimize } -1 + [-1 \ \ -1]\begin{bmatrix} d_1 \\ d_2 \end{bmatrix} = -1 - d_1 - d_2 = -x_1 - x_2 + 1$$

$$\text{Subject to} -\frac{2}{3} + \begin{bmatrix}\frac{1}{3} & \frac{1}{3}\end{bmatrix}\begin{bmatrix} d_1 \\ d_2 \end{bmatrix} = \frac{1}{3}(d_1 + d_2) - \frac{2}{3} = \frac{1}{3}(x_1 + x_2 - 4) \leq 0 \qquad \text{and}$$

$$-1 + [-1 \ \ 0]\begin{bmatrix} d_1 \\ d_2 \end{bmatrix} = -d_1 - 1 = -x_1 \leq 0 \qquad \text{and}$$

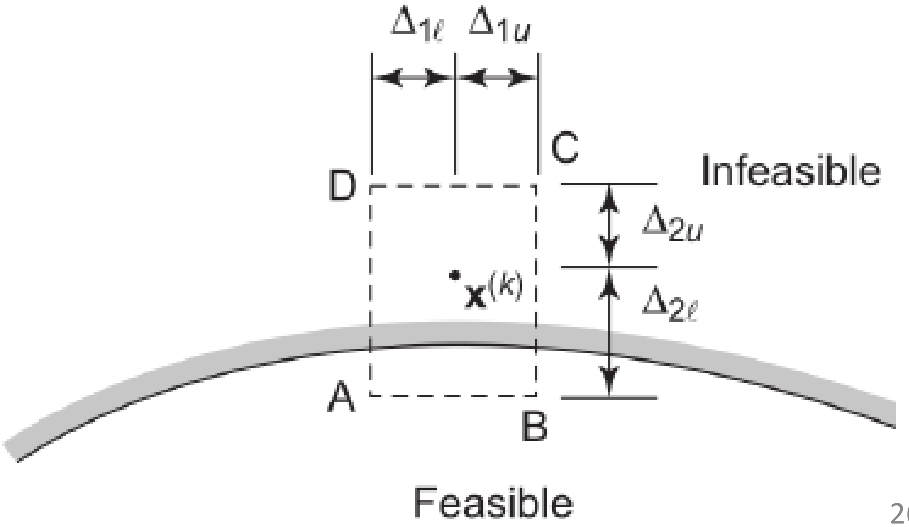
$$-1 + [0 \ \ -1]\begin{bmatrix} d_1 \\ d_2 \end{bmatrix} = -d_2 - 1 = -x_2 \leq 0$$





Sequential Linear Programming

- Since the sub-problem is linear in each iteration, it is easy to solve (e.g., using the Simplex method).
- The linearized function only fit the real problem in a small neighborhood around the current point.
- Move limits are imposed on the sub-problem to not make too large moves.
- The limits may be changed in each iteration.
- The limits will ensure that the linear sub-problem is bounded.
- Since this limits the step size, a line search is not used.



Sequential Linear Programming

$$\min_{\boldsymbol{x}} f(\boldsymbol{x}) \text{ such that}$$
$$h_i(\boldsymbol{x}) = 0 \ , \ i = 1, ..., p$$
$$g_j(\boldsymbol{x}) \leq 0 \ , \ j = 1, ..., m$$

Set a maximum constraint violation ϵ_1, a minimum move ϵ_2 and a maximum number of steps k_{max}.
Choose an inital design $\boldsymbol{x}^0$.
$k = 0$.
While ($k = 0$ or $g_i > \epsilon_1$ or $|h_i| > \epsilon_1$ or $|\boldsymbol{d}|^k > \epsilon_2$) and $k < k_{max}$:
- Linearize the problem around $\boldsymbol{x}^k$.
- Select move limits Δ_{il}^k and Δ_{iu}^k for all design variables.
- Solve the linear programming subproblem augmented with move limits to obtain $\boldsymbol{d}^k$.
- $\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + \boldsymbol{d}^k$.
- $k = k + 1$.

The linear sub-problem can be solved with special methods for linear problems (e.g. the Simplex method).

The hard part is to choose appropiate move limits that are not too loose and not too tight.

Sequential Linear Programming works well for improving a design, but should not be used as a general "black-box" solver for non-linear constrained optimization problems due to the deficiencies highlighted above.

Next time we discuss sequential quadratic programming and the method of moving asymptotes. Both use some of the ideas from sequential linear programming and works extremely well.

We will mainly use the method of moving asymptotres in this course.